IVF Clinic Management System

Lab Technician Role — Full Software Engineering Deliverable

Silvja Fejzulla | Software Engineering | 2026

1. User Scenarios List

The table below lists all key scenarios in which the Lab Technician interacts with the IVF Clinic Management System. Each scenario is a realistic narrative of a task the user needs to accomplish.

ID	Scenario	Actor	Summary	Priority
US-S01	Maja urgently needs a result	Lab Technician	Maja's doctor orders a hormone panel. The lab technician opens the test queue, sees it flagged as urgent, processes the sample, uploads the PDF report, and the system auto-flags a critically low FSH value and immediately alerts the doctor.	High
US-S02	Recording day-3 embryo observations	Lab Technician	After morning observation, the technician opens the embryology module for Arta's cycle, enters day-3 data: 8 cells, 10% fragmentation, grade BB. Optionally attaches a microscopy image. The system saves the entry and the doctor sees the update on their dashboard.	High
US-S03	Freezing an embryo for FET	Lab Technician	A good-quality blastocyst is to be frozen. The technician selects 'Freeze Embryo', enters tank C4, cane 2, straw 3, vitrification method, and today's date. The embryo status updates to Frozen and is linked to Arta's FET plan.	High
US-S04	Receiving a donated sperm sample	Lab Technician	A new donor sample arrives. The technician scans the barcode, enters donor ID, sample type (sperm), quantity, collection date, and storage location. The sample is logged as 'Received' with a chain-of-custody entry and placed in the bank with status Pending.	High
US-S05	Clearing a donor sample after screening	Lab Technician	Screening results arrive for donor D-042. The technician opens the donor record, changes status from Pending to Cleared, adds a mandatory reason note. The sample becomes eligible for patient assignment.	High
US-S06	Tracking a sample through processing	Lab Technician	The technician scans sample LB-2091, sees the current custody log (Received → Stored), selects 'Processed', confirms. The system logs the event with timestamp and technician ID. Full traceability is maintained.	Medium

ID	Scenario	Actor	Summary	Priority
US-S07	Logging a failed QC check	Lab Technician	During morning QC, the hormone analyser fails the calibration check. The technician records QC: Fail, corrective action: 'Reagent lot R-22 replaced'. The system flags the instrument as 'Pending QC clearance' and blocks it from patient testing until a Pass is entered.	High
US-S08	Scheduling calibration for an overdue instrument	Lab Technician	The technician notices the centrifuge calibration is 3 days overdue. They open the equipment record, update the calibration date, set status to Active, add a note. The system records the event and clears the overdue flag.	Medium
US-S09	Receiving a low-stock alert	Lab Technician / Hospital Admin	The cryoprotectant reagent drops below the configured minimum. The system sends an alert to the lab technician and hospital administrator. The technician initiates a restocking request before operations are disrupted.	Medium
US-S10	Generating the daily lab summary	Lab Technician	At end of shift, the technician generates the daily activity report: 12 tests processed, 3 pending, 1 result flagged, 1 equipment issue. The summary is available for the lab supervisor.	Low

2. User Scenarios — Extended

Each scenario below is expanded with preconditions, step-by-step flow, alternatives, and postconditions following the structured scenario template.

US-S01 — Urgent test result — auto-flag

Actor	Lab Technician
Goal	Process an urgent test request and ensure abnormal results are immediately communicated to the doctor.
Preconditions	Technician is authenticated. A doctor-created test order is marked urgent in the queue.

Main flow	1. Open test queue — urgent order is at the top. 2. Select the order, view patient and doctor details. 3. Process sample and upload result PDF. 4. System compares values against reference ranges. 5. FSH value is critically low — system auto-flags and sends urgent alert to doctor. 6. Audit log records upload: technician ID, timestamp, order reference.
Alternative / Exception	If the upload file exceeds 25 MB, the system rejects it with an inline error. If the doctor order is not found, the technician searches by patient ID.
Postconditions	Report stored and linked to patient record. Doctor notified. Urgent alert sent. Audit log updated.
Frequency	Daily, multiple times.

US-S02 — Record day-3 embryo observations

Actor	Lab Technician
Goal	Record accurate daily embryo development data to support the medical team's quality assessment.
Preconditions	Technician is authenticated and assigned to Arta's active IVF cycle. Fertilised embryos are registered.
Main flow	1. Navigate to Embryology Module. 2. Search for and select Arta's active cycle. 3. Select the embryo to update. 4. Enter day 3 data: 8 cells, 10% fragmentation, grade BB, free-text note. 5. Optionally attach microscopy image. 6. System validates mandatory fields and saves as immutable entry. 7. Doctor and patient dashboards update in near-real-time.
Alternative / Exception	If a mandatory field (day, cell count, grade) is missing, the system blocks submission. If the microscopy image exceeds 10 MB, it is rejected with an error.
Postconditions	Embryo record updated. Entry is immutable. Doctor sees new data on dashboard.
Frequency	Daily during the embryo culture period (typically days 1–6).

US-S03 — Freeze a blastocyst for FET

Actor	Lab Technician
Goal	Record cryopreservation details for a good-quality blastocyst to support future frozen embryo transfer.
Preconditions	Technician is authenticated. The embryo record exists and the doctor has added a freeze instruction.

Main flow	1. Open embryo record in the Embryology Module. 2. Confirm the doctor's freeze instruction is displayed. 3. Select 'Freeze Embryo'. 4. Enter: tank C4, cane 2, straw 3, vitrification method, today's date. 5. System saves cryopreservation record and updates embryo status to Frozen.
Alternative / Exception	If storage location fields are incomplete, the system blocks submission. If cycle is cancelled by doctor, technician can add a final note but cannot freeze.
Postconditions	Embryo status set to Frozen. Cryopreservation record linked to embryo. Audit log updated.
Frequency	As required per IVF cycle.

US-S04 — Receive a donated sperm sample

Actor	Lab Technician
Goal	Register a new donor sample into the donation bank with full chain-of-custody from arrival.
Preconditions	Technician is authenticated. Donor D-055 is registered in the system.
Main flow	1. Navigate to Donation Bank module. 2. Select 'Add New Sample'. 3. Scan barcode or enter donor ID, sample type (sperm), quantity, collection date, storage location. 4. System validates donor ID against donor records. 5. System adds sample with status Pending and logs a 'Received' custody event.
Alternative / Exception	If donor ID is not found, the system prompts to register the donor first before adding the sample.
Postconditions	Sample added to inventory. Chain-of-custody 'Received' event logged. Sample status is Pending.
Frequency	As required when donor samples arrive.

US-S05 — Update donor screening status to Cleared

Actor	Lab Technician
Goal	Mark a screened donor sample as Cleared so it becomes eligible for patient treatment.
Preconditions	Technician is authenticated. Donor sample exists with status Pending.
Main flow	1. Open Donation Bank module, find donor D-042. 2. Select 'Update Screening Status'. 3. Change status from Pending to Cleared. 4. Enter mandatory reason note. 5. System saves the change and makes sample available for assignment.

Alternative / Exception	If the mandatory reason note is missing, submission is blocked. If status is set to Rejected, sample is immediately removed from the assignable pool.
Postconditions	Sample status updated. Change logged with technician ID and timestamp. Sample available for patient assignment if Cleared.
Frequency	Following completion of donor screening.

US-S06 — Log a sample transfer event

Actor	Lab Technician
Goal	Maintain full chain-of-custody by logging a sample transfer to another department.
Preconditions	Technician is authenticated. Sample LB-2091 is registered in the system.
Main flow	1. Navigate to Sample Tracking module. 2. Scan barcode or enter sample ID LB-2091. 3. System displays current custody history. 4. Select event type: Transferred. 5. Enter recipient name and role. 6. System records event with timestamp and technician ID.
Alternative / Exception	If sample ID is not found, the system returns an error and prompts to verify or register the sample.
Postconditions	Transfer event recorded as immutable entry in custody log. Full history available for audit.
Frequency	As required during sample handling.

US-S07 — Record a failed QC check and corrective action

Actor	Lab Technician
Goal	Document a failed instrument QC check and the corrective action taken to restore compliance.
Preconditions	Technician is authenticated. Equipment records are configured in the Equipment Management module.
Main flow	1. Navigate to Equipment Management module. 2. Select the hormone analyser. 3. Select 'Record QC Check'. 4. Enter QC result values, select Fail. 5. Enter corrective action: 'Reagent lot R-22 replaced'. 6. System saves QC record and flags instrument as 'Pending QC clearance'. 7. System blocks instrument from patient testing until a Pass QC is entered.
Alternative / Exception	If corrective action field is left empty on a Fail result, submission is blocked.
Postconditions	QC record saved. Instrument flagged and blocked from patient testing. Audit log updated.
Frequency	At least once per shift or as required by QC protocol.

US-S08 — Update overdue equipment calibration

Actor	Lab Technician
Goal	Update the calibration record for an overdue instrument to restore it to compliant active status.
Preconditions	Technician is authenticated. The centrifuge record shows calibration as overdue.
Main flow	1. Navigate to Equipment Management module. 2. Select the centrifuge (status: Calibration Required). 3. Update calibration date and maintenance status to Active. 4. Add a note. 5. System saves the update with timestamp and technician ID. Overdue flag cleared.
Alternative / Exception	If the instrument has a failed QC in addition to overdue calibration, both issues must be resolved before the instrument is released for patient testing.
Postconditions	Equipment record updated. Overdue flag cleared. Instrument available for patient testing.
Frequency	As required per calibration schedule.

3. Use Cases

Use cases follow the structured template format used in the course. Each use case maps to one or more user stories from the Lab Technician role.

UC_LT_01: Upload Test Report

Field	Detail
Use Case ID	UC_LT_01
Use Case Name	Upload Test Report
Actor(s)	Lab Technician (primary), Doctor (notified), Patient (indirect via portal)
Preconditions	Lab Technician is authenticated. A lab test order exists (created by a doctor). Test has been performed and results are ready.

Field	Detail
Main Success Flow	1. Technician navigates to the Test Queue and views pending orders sorted by urgency. 2. Technician selects the relevant test order. 3. System displays order details: patient name, test type, requesting doctor, clinical notes. 4. Technician uploads result file (PDF/CSV/image) or enters values manually. 5. Technician reviews before submission. 6. System compares result values against predefined reference ranges. 7. System saves report linked to patient record and doctor's order. 8. System timestamps the entry and records technician ID. 9. System notifies requesting doctor that results are ready. 10. Doctor reviews and approves visibility; patient can then view in portal.
Alternative / Exception Flows	4a. Invalid/out-of-range entry: system highlights and requires confirmation. 6a. Abnormal result: system auto-flags with colour indicator and sends urgent alert to doctor. 2a. Order not found: technician searches by patient ID or doctor name.
Postconditions	Test report stored and linked to patient record. Doctor notified. Urgent alert sent if flagged. Audit log records: technician ID, timestamp, order reference.
Constraints & Restrictions	Only authenticated lab technicians may upload results. Results must be linked to an existing doctor order. Max file size 25 MB; accepted formats: PDF, JPEG, PNG, CSV. Results cannot be deleted — corrections require an amended report with reason. Patient access is gated by doctor review.

UC_LT_02: Record Embryo Development & Cryopreservation

Field	Detail
Use Case ID	UC_LT_02
Use Case Name	Record Embryo Development & Cryopreservation
Actor(s)	Lab Technician (primary), Doctor (views and adds instructions), Patient (read-only view)
Preconditions	Lab Technician is authenticated. An active IVF cycle exists with fertilised embryos. Technician is assigned to this patient's cycle.

Field	Detail
Main Success Flow	1. Technician navigates to the Embryology Module. 2. Searches for and selects the patient's active cycle. 3. System displays embryo list with latest data and any doctor instructions. 4. Technician selects an embryo to update. 5. Enters: development day, cell count, fragmentation %, morphology grade, free-text notes. 6. Optionally uploads microscopy image (max 10 MB). 7. System validates mandatory fields. 8. System saves entry as immutable with timestamp and technician ID. 9. Doctor and patient dashboards update in near-real-time. 10. For cryopreservation: technician selects 'Freeze Embryo', enters tank, cane, straw, date, vitrification method. 11. System records cryopreservation event and updates embryo status to Frozen.
Alternative / Exception Flows	3a. No embryos recorded: system prompts for day-0 fertilisation data. 5a. Value outside expected range: system warns but allows entry with mandatory confirmation note. 7a. Mandatory field missing: system blocks submission. 6a. Cycle cancelled by doctor: technician can add final note but cannot transition cycle.
Postconditions	Embryo record updated as immutable entry. Doctor and patient views reflect new data. If cryopreserved, embryo status = Frozen and storage location recorded.
Constraints & Restrictions	Only assigned lab technicians may record embryo data. Entries are append-only. Minimum mandatory fields: day, cell count, morphology grade. Cryopreservation records must include tank, cane, straw, date, method. Patient data is role-restricted.

UC_LT_03: Manage Donation Bank Stock & Donor Sample Status

Field	Detail
Use Case ID	UC_LT_03
Use Case Name	Manage Donation Bank Stock & Donor Sample Status
Actor(s)	Lab Technician (primary), Hospital Administrator (receives low-stock alerts)
Preconditions	Lab Technician is authenticated. Donation Bank module is configured with at least one donor record.

Field	Detail
Main Success Flow	1. Technician navigates to Donation Bank module. 2. System displays current inventory: type, donor ID, quantity, collection date, expiry date, storage location, screening status. 3. Technician selects action: Add New Sample / Update Quantity / Update Screening Status / Mark as Expired. 4. Add New Sample: enters donor ID, type, quantity, collection date, storage location, status (Pending). 5. System validates donor ID and logs a 'Received' custody event. 6. Update Screening Status: technician sets status to Cleared or Rejected with mandatory reason note. 7. Rejected samples are immediately removed from the assignable pool. 8. System sends low-stock alert if any sample type falls below configured minimum threshold.
Alternative / Exception Flows	4a. Donor ID not found: system prompts to register donor first. 8a. Quantity below threshold: system immediately triggers low-stock notification to administrator. Expiry path: system flags samples expiring within 30 days; expired samples locked from assignment.
Postconditions	Inventory updated. All changes logged as custody events with technician ID and timestamp. Rejected/expired samples removed from assignable pool. Low-stock alert sent if threshold breached.
Constraints & Restrictions	All donor samples must be linked to a registered donor. Expired or rejected samples cannot be assigned to patients. Every inventory change requires authentication and is permanently logged.

UC_LT_04: Log Sample Chain-of-Custody Event

Field	Detail
Use Case ID	UC_LT_04
Use Case Name	Log Sample Chain-of-Custody Event
Actor(s)	Lab Technician (primary)
Preconditions	Lab Technician is authenticated. A patient sample or donor sample record exists in the system.

Field	Detail
Main Success Flow	1. Technician navigates to Sample Tracking module. 2. Scans barcode or manually enters sample ID. 3. System retrieves and displays the sample's current custody history. 4. Technician selects event type: Received / Processed / Stored / Transferred / Discarded. 5. For Transferred: technician enters destination (name and role/department). 6. For Discarded: technician enters disposal reason and method. 7. System records the event: event type, timestamp, technician ID, additional notes. 8. System displays updated chain-of-custody log.
Alternative / Exception Flows	2a. Sample ID not found: system returns error; technician must verify or register sample first. 4a. Discarded event: system requires mandatory reason code and disposal confirmation before saving.
Postconditions	New chain-of-custody entry created and linked to sample record. Entry is immutable. Full custody history available for regulatory audit.
Constraints & Restrictions	All custody events must be linked to an authenticated technician. Event records are immutable. Disposal events require mandatory reason code from a controlled list. Transfer events must record recipient identity. Logs must be retained per local regulatory requirements.

UC_LT_05: Manage Lab Equipment Records & Perform QC Checks

Field	Detail
Use Case ID	UC_LT_05
Use Case Name	Manage Lab Equipment Records & Perform QC Checks
Actor(s)	Lab Technician (primary)
Preconditions	Lab Technician is authenticated. Equipment records are registered in the Equipment Management module. QC criteria are configured in the system.

Field	Detail
Main Success Flow	1. Technician navigates to Equipment Management module. 2. System displays equipment list: name, last calibration date, next calibration due, status. 3. Technician selects an instrument to update. 4. Updates calibration date or maintenance status with a note. 5. System saves update with timestamp and technician ID. 6. System highlights instruments overdue or due within 7 days. 7. For QC check: technician selects 'Record QC Check' for instrument or reagent lot. 8. Enters QC result values and selects Pass or Fail. 9. If Fail: enters corrective action taken. 10. System saves QC record with timestamp, technician ID, and corrective action. 11. System displays warning if instrument has failed QC and is not yet cleared; blocks from patient testing.
Alternative / Exception Flows	6a. Calibration overdue: system changes status to 'Calibration Required'; instrument blocked from patient tests until recalibrated. 8a. QC Fail: system blocks instrument or reagent lot from patient samples until a Pass QC is entered. 3a. Equipment record not found: technician can create a new record.
Postconditions	Equipment record updated with latest calibration, status, and QC results. Instruments with failed QC or overdue calibration flagged and blocked. Corrective actions documented for audit.
Constraints & Restrictions	Only authenticated technicians may update equipment or QC records. Overdue or failed-QC instruments cannot be used for patient testing until cleared. QC checks must reference a specific reagent lot or instrument serial number. All records are immutable once saved. Calibration intervals and QC thresholds are set by the lab manager.

4. Functional Requirements

Functional requirements describe what the system must do for the Lab Technician role. Each requirement is linked to the use case it supports and assigned a priority.

Req ID	Requirement	Related UC	Priority
FR-LT-01	The system shall display a prioritised queue of pending lab test orders sorted by urgency level and due time.	UC_LT_01	High

Req ID	Requirement	Related UC	Priorit y
FR-LT-02	The system shall allow the lab technician to receive and label patient samples via barcode scan or manual sample ID entry.	UC_LT_01	High
FR-LT-03	The system shall allow upload of test result files (PDF, JPEG, PNG, CSV; max 25 MB) linked to the patient record and the originating doctor order.	UC_LT_01	High
FR-LT-04	The system shall automatically compare uploaded result values against predefined clinical reference ranges and flag any abnormal values with a colour-coded indicator.	UC_LT_01	High
FR-LT-05	The system shall send an urgent alert to the requesting doctor whenever an abnormal result is flagged.	UC_LT_01	High
FR-LT-06	The system shall notify the requesting doctor when a test report is uploaded and ready for review.	UC_LT_01	High
FR-LT-07	The system shall allow the lab technician to record daily embryo development observations including development day, cell count, fragmentation percentage, morphology grade, and free-text notes.	UC_LT_02	High
FR-LT-08	The system shall allow optional attachment of a microscopy image (max 10 MB) to each embryo development entry.	UC_LT_02	Mediu m
FR-LT-09	The system shall allow the lab technician to record cryopreservation details: storage tank, cane, straw position, freezing date, and vitrification method.	UC_LT_02	High
FR-LT-10	The system shall update the embryo status to 'Frozen' upon successful cryopreservation record entry.	UC_LT_02	High
FR-LT-11	The system shall display any clinical notes or freeze/transfer instructions added by the doctor to the embryo record.	UC_LT_02	Mediu m
FR-LT-12	The system shall allow the lab technician to add, update, and view donation bank stock (sperm, egg, embryo) including quantity, collection date, expiry date, and storage location.	UC_LT_03	High
FR-LT-13	The system shall allow the lab technician to update a donor sample's screening status (Pending → Cleared / Rejected) with a mandatory reason note.	UC_LT_03	High
FR-LT-14	The system shall automatically remove Rejected or expired samples from the assignable pool.	UC_LT_03	High
FR-LT-15	The system shall send a low-stock alert to the lab technician and hospital administrator when any sample type falls below the configured minimum threshold.	UC_LT_03	Mediu m
FR-LT-16	The system shall allow the lab technician to log chain-of-custody events (Received, Processed, Stored, Transferred, Discarded) linked to a sample by barcode or ID.	UC_LT_04	Mediu m
FR-LT-17	The system shall require a mandatory reason code for Discarded events and record the recipient identity for Transferred events.	UC_LT_04	Mediu m
FR-LT-18	The system shall store all custody events as immutable records with timestamp and technician ID.	UC_LT_04	High

Req ID	Requirement	Related UC	Priority
FR-LT-19	The system shall allow the lab technician to view and update equipment records including instrument name, last calibration date, next calibration due date, and status.	UC_LT_05	Medium
FR-LT-20	The system shall highlight instruments whose next calibration date is within 7 days or overdue.	UC_LT_05	Medium
FR-LT-21	The system shall allow the lab technician to record QC checks (Pass/Fail) per instrument or reagent lot, with mandatory corrective action note on Fail.	UC_LT_05	High
FR-LT-22	The system shall block the use of any instrument with a failed QC or overdue calibration from patient testing until cleared.	UC_LT_05	High
FR-LT-23	The system shall allow the lab technician to schedule and record preventive maintenance tasks with date and technician responsible.	UC_LT_05	Medium
FR-LT-24	The system shall generate a daily lab activity summary report (tests processed, pending, flagged, equipment issues) viewable by the lab supervisor.	US_LT_15	Low

5. Non-Functional Requirements

Non-functional requirements define the quality attributes, constraints, and standards the system must meet to support the Lab Technician role effectively and safely.

Req ID	Category	Requirement	Priority
NFR-LT-01	Performance	Test report upload (up to 25 MB) shall complete within 5 seconds on a standard hospital network connection.	High
NFR-LT-02	Performance	The test queue page shall load and display up to 100 pending orders within 2 seconds.	High
NFR-LT-03	Reliability	The system shall maintain 99.9% uptime during clinic operating hours (06:00–22:00).	High
NFR-LT-04	Security	All lab technician actions (uploads, inventory changes, custody events) shall require authenticated sessions with role-based access control.	High
NFR-LT-05	Security	Audit logs shall be immutable and tamper-evident; all entries must include technician ID, timestamp, and the record modified.	High
NFR-LT-06	Security	The system shall comply with applicable data-protection regulations (e.g. GDPR) for all patient and donor records accessed by the lab technician.	High

Req ID	Category	Requirement	Priority
NFR-LT-07	Usability	The test queue, embryology module, and donation bank screens shall be navigable within 3 clicks from the lab technician dashboard.	Medium
NFR-LT-08	Usability	Error messages for invalid field values shall appear inline next to the field within 500 ms and clearly state what is wrong.	Medium
NFR-LT-09	Scalability	The system shall support up to 500 concurrent lab technician sessions without degradation of response times.	Medium
NFR-LT-10	Maintainability	Clinical reference ranges and QC pass/fail thresholds shall be configurable by the lab manager without code deployment.	High
NFR-LT-11	Compliance	All QC records and equipment calibration logs shall be retained for the minimum period required by ISO 15189 and local regulatory authority.	High
NFR-LT-12	Interoperability	The API shall expose RESTful endpoints for all lab technician features, accepting and returning JSON, to allow integration with future front-end clients or third-party LIS.	High
NFR-LT-13	Traceability	All embryo development entries shall be append-only; once confirmed they cannot be deleted or edited — corrections must be new entries referencing the original.	High
NFR-LT-14	Availability	Urgent abnormal-result alerts shall be delivered to the requesting doctor within 30 seconds of the lab technician submitting the result.	High

End of Lab Technician Deliverable — Silvja Fejzulla | 2026